Biospherical Instruments Inc.

CALIBRATION CERTIFICATE

Calibration Date	03/24/14				
Model Number	QSP2155		Firmware Version		v 1.2
Serial Number	50182				
Operator	TPC				
Standard Lamp	V-033(3/7/12)				
Sensor Operating Voltage Range:		6	to	15	VDC (+)
Output Polarity:	Positive				

Sensor Output Voltage:

Sensor Illuminated	165.6	mV
✓Sensor Dark	10.6	mV
Sensor Net Response	154.9	mV
RG780	10.6	mV

Corrected Lamp Output:

Output In Air (same condition as calibration):

1.55E-02 uE/cm²sec

Output Corrected for Immersion in Water: Using Immersion Coefficient of:

2.74E-02 uE/cm²sec 0.5664 (Collector Type: SC-3)

Calibration Scale Factor:

(To calculate irradiance, divide the net voltage reading in Volts by this value.)

✓Dry:	9.988E+00	Volts/(**uE/cm²sec**)
Wet:	5.657E+00	Volts/(**uE/cm²sec**)
Dry:	9.988E-04	Volts/(**uE/m²sec**)
Wet:	5.657E-04	Volts/(**uE/m²sec**)

Notes:

1. Annual calibration is recommended.
2. Calibration is performed using a Standard of Spectral Irradiance traceable to the National Institute of Standards and Technology (NIST).
3. To approximate the sensor's saturating irradiance, multiply the calibration factor by the sensor power supply voltage, minus one volt.
4. The collector should be cleaned frequently with alcohol.